

Machine Learning for Solid-State Battery Materials Discovery

Abstract

Solid-state batteries are among the most intensively studied next-generation electrochemical energy-storage technologies because they promise improved safety, compatibility with lithium metal, and potentially higher cell-level energy density than conventional liquid-electrolyte lithium-ion batteries. At the same time, their materials design problem is unusually difficult: a practically useful solid-state battery requires not only a fast ion conductor, but also electrochemical and chemical stability against electrodes, manufacturable interfaces, sufficient mechanical robustness, favorable microstructure evolution, and acceptable cost and synthesizability. Over the past decade, machine learning has reshaped how this design space is navigated. The field has progressed from descriptor-based screening of candidate solid electrolytes to graph neural networks, transformer-style composition models, generative crystal design, machine-learning interatomic potentials, literature mining, computer-vision analysis of operando data, and early forms of closed-loop experimental optimization [1–10,17–18,38–68,87–128].

This survey argues that the most important conceptual shift has been from “predicting a single scalar property” to building multi-fidelity, uncertainty-aware, and increasingly autonomous discovery pipelines. The strongest results have so far appeared in inorganic solid electrolytes, especially Li- and Na-ion conductors, where curated conductivity labels and structure databases have enabled high-throughput screening. However, the central bottlenecks of practical solid-state batteries now lie at the system boundary: interfaces, composite electrodes, chemo-mechanics, and synthesis-sensitive microstructure. Consequently, the most promising future direction is not merely larger machine-learning models, but tighter integration among literature extraction, database curation, crystal and composition generation, DFT and ML-potential validation, interface modeling, and experiment-in-the-loop optimization [8–10,14–16,45–52,57–73,87–128].

Introduction

Solid-state battery research sits at the intersection of materials discovery, electrochemistry, mechanics, and manufacturing science. Inorganic and polymer solid electrolytes can, in principle, replace flammable liquids and enable the use of lithium metal, thereby expanding the design space for safe, high-energy cells. But the putative advantages of solid-state architectures depend on simultaneous success across multiple materials classes: the electrolyte must conduct ions rapidly, the electrode–electrolyte interfaces must remain stable, the composite cathode must sustain coupled ionic/electronic transport under realistic loadings, and the full cell must tolerate chemo-mechanical stress over long cycling [1–9,108–118,127–150]. This coupling is precisely what makes solid-state battery discovery a compelling application domain for machine learning.

From the standpoint of materials informatics, solid-state batteries are also a revealing stress test. Some target quantities, such as formation energy, band gap, elastic constants, or crystal structure prototypes, are abundant in high-throughput computational repositories. Others, such as room-temperature ionic conductivity, activation energy, interfacial resistance, critical current density, or lifetime under stack

pressure, are scarce, noisy, and heavily conditioned on sample history and measurement protocol [3–5,8–10,38–50,128–150]. The field therefore contains nearly every important machine-learning challenge in modern scientific discovery: small and biased datasets, multi-fidelity supervision, out-of-distribution generalization, physically constrained extrapolation, uncertainty quantification, and the need to couple screening with expensive mechanistic validation.

The literature has evolved in roughly three phases. The first phase used physics-based heuristics and high-throughput first-principles screening to identify plausible ionic conductors and chemically stable chemistries [149,150]. The second phase introduced supervised machine learning, usually with hand-engineered descriptors, to accelerate conductivity and stability prediction and to pre-screen huge structural databases before expensive atomistic calculations [17–25,32–37,74–86]. The third phase, which is still unfolding, has moved toward learned representations, generative design, machine-learning interatomic potentials, multimodal data extraction, computer vision for operando characterization, and real-time optimization of cycling or synthesis decisions [45–68,87–128].

This survey focuses on that third phase while retaining enough physical and computational context to clarify why certain method families have succeeded and why others remain immature. The emphasis is on materials discovery rather than battery management alone, but the scope is intentionally broader than solid-electrolyte screening in the narrow sense. In practice, the discovery problem spans electrolyte bulk chemistry, defects and disorder, interfaces and interphases, cathode and anode compatibility, and increasingly the microstructural and process variables that determine whether a nominally promising material actually performs in an all-solid-state cell [7–9,16,108–118,127–150].

Methods

A useful way to organize the methodological landscape is by the point in the discovery pipeline where machine learning enters: data construction, surrogate property prediction, learned representation and generation, atomistic simulation acceleration, and experiment-coupled inference.

Data foundations and task formulation.

The first prerequisite for machine learning in solid-state batteries is a data substrate that is both large enough to train useful models and structured enough to preserve scientific meaning. For solid-state materials discovery, this substrate has come from two complementary sources. The first is high-throughput computational repositories, including Materials Project, AFLOW, OQMD, JARVIS, Materials Cloud, and more recent large-scale crystal databases such as GNoME and Alexandria, which provide structural, thermodynamic, and electronic descriptors at scales that make representation learning feasible [38–44]. The second is smaller but far more decision-relevant collections of experimentally measured transport data, such as curated lithium-ion conductor datasets and the recent OBELiX benchmark, together with text-mined datasets extracted from articles by natural-language-processing pipelines and, more recently, large-language-model-based systems [45–55,147,148].

These data sources define distinct supervised tasks. Computational databases support proxy-learning tasks such as phase stability, band gap, oxidation/reduction propensity, elastic modulus, structural classification, migration-barrier surrogates, and defect energetics. Curated experimental datasets support more application-facing tasks, notably room-temperature ionic conductivity, Arrhenius activation energy, electrochemical window, and sometimes synthesis-condition prediction. The tension between them is fundamental: the easiest labels to compute are not always the ones that determine cell-level performance,

whereas the most practical labels are often sparse, inconsistent, and confounded by grain boundaries, pellet density, processing history, dopant inhomogeneity, and measurement conventions [8–10,14–16,45–52,128–150]. In solid-state batteries, model design is therefore inseparable from target design.

Supervised property prediction and high-throughput screening.

Classical machine learning entered the field through the promise of replacing enormous candidate lists with ranked shortlists. Random forests, gradient boosting, support-vector machines, Gaussian-process models, and shallow neural networks were applied to descriptors derived from composition, ionic radii, electronegativity, bond valence, crystal symmetry, lattice geometry, and simple electronic proxies. Early landmark studies used these models to screen lithium conductors with substantially greater throughput than direct DFT or ab initio molecular dynamics, and later work extended the approach to garnet, NASICON, antiperovskite, argyrodite, and sodium-conductor subspaces [17–37,74–86].

This family of methods remains important because the small-data regime is still common in solid-state battery discovery. Descriptor-based models can be surprisingly strong when coupled with expert curation and physically meaningful feature engineering. For example, conductivity classifiers or regressors trained on family-specific datasets often identify chemically plausible candidates at low computational cost and offer interpretable feature rankings, which is useful when a domain expert wants to understand whether the model is simply learning family labels or discovering transferable transport trends [20–37]. Hybrid pipelines that combine classical machine learning with bond-valence analysis, prototype substitution, or post hoc DFT filtering have been especially effective because they place machine learning where it is strongest—cheap prioritization—while leaving final validation to more reliable physics-based methods [20–25,32–37,79].

Yet the limitations of this paradigm are equally clear. Many conductivity datasets are composition-heavy and mechanism-light. A model may distinguish “good argyrodites” from “bad garnets” without learning why a specific bottleneck, defect landscape, or anion sublattice disorder pattern promotes conduction. Likewise, a conductivity label obtained from a dense hot-pressed pellet is not commensurate with one measured on a poorly sintered specimen, even when the nominal composition is identical. This means reported accuracy can exaggerate real generalization, especially when train and test sets share closely related families or dopant series [8–10,14–16,20–37,45–47].

Deep representation learning and generative design.

The rise of deep learning changed this picture by reducing reliance on hand-crafted descriptors and enabling models to operate directly on crystal graphs, compositions, or atomistic environments. Crystal Graph Convolutional Neural Networks, MEGNet, M3GNet, ALIGNN, SchNet-derived architectures, CrabNet, and other transformer-style or message-passing models have all been adopted in adjacent materials-informatics tasks and, increasingly, in solid-state battery-specific pipelines [57–65]. Their main attraction is representational flexibility: graph models can encode local coordination and topology, while composition-only transformers offer rapid screening when crystal structures are unavailable or uncertain.

In solid-state battery applications, these models have been used to predict conductivity-related quantities, mechanical properties relevant to dendrite suppression, and broad materials properties that serve as constraints in multi-objective screening. Composition-based models are especially useful in the earliest discovery stage, where the question is whether a new stoichiometry is even worth structural relaxation or synthesis; graph-based models become more valuable once a candidate structure is known or can be generated. The practical lesson from the literature is not that one paradigm dominates, but that model

choice should track information availability. Composition-only models are robust for rapid exploration and literature-derived datasets; structure-aware models are preferable when local pathways, bottlenecks, or topology-sensitive stability are central [20–37,57–65,74–86].

Generative design extends this logic from ranking existing candidates to proposing new ones. Variational autoencoders, diffusion-based crystal generators, inverse-design frameworks, and evolutionary or graph-based generative methods are now able to generate stable or metastable crystal candidates, sometimes conditioned on target properties or structure classes [66–73]. In principle, this is a natural fit to solid-state battery discovery, where known chemistry families cover only a small part of compositional space. In practice, however, generative design remains constrained by three problems. First, a generated crystal must be stable enough to matter, which usually requires a second validation layer. Second, even a stable crystal may be irrelevant if it has poor transport or catastrophic interface chemistry. Third, the space of synthesizable and processable solid-state battery materials is narrower than the space of mathematically valid crystals. The most effective generative pipelines therefore do not generate once and declare success; they alternate generation, surrogate evaluation, uncertainty estimation, and physics-based filtering [62,66–73].

Recent literature-mining and language-model work suggests a complementary route to generation: not generating new crystals from scratch, but generating structured hypotheses from the scientific record. Large-language-model systems can now extract composition, phase, morphology, processing conditions, and application relations from articles at scales that were previously impractical, helping transform narrative literature into machine-readable prior knowledge [51–56,147,148]. For solid-state batteries, this is especially valuable because failure modes, processing caveats, and interface-specific observations are often more richly described in prose than in databases.

Machine-learning interatomic potentials and multiscale atomistic discovery.

Perhaps the most consequential methodological advance for solid-state battery materials discovery has been the emergence of machine-learning interatomic potentials. Unlike scalar property predictors, ML potentials approximate the full high-dimensional potential-energy surface and therefore unlock molecular-dynamics simulations at spatial and temporal scales far beyond standard *ab initio* molecular dynamics while retaining near-DFT accuracy when carefully trained [87–107]. This matters profoundly for solid-state batteries because ionic conductivity, defect-assisted diffusion, correlated motion, structural transitions, and many interfacial reactions are kinetic phenomena that cannot be inferred reliably from composition alone.

The impact is visible in at least three kinds of study. First, ML potentials can reproduce and reinterpret transport mechanisms in canonical solid electrolytes, revealing when simple activation-barrier pictures fail or when collective motion and phase transitions dominate observed conductivity [91–103]. Second, they enable active-learning loops in which expensive quantum calculations are targeted only to uncertain regions of configuration space, making large-scale simulations of defects, disorder, grain boundaries, and interfaces tractable [88–103]. Third, ML potentials increasingly function as discovery engines in their own right: they can compute diffusivities, estimate transport-relevant descriptors, and screen broad materials spaces before final DFT confirmation [74–79,91–107].

This approach narrows the gap between “screening” and “mechanism.” A classical surrogate might predict that a candidate belongs to the high-conductivity class; an ML potential can explain whether that conductivity arises from frustration, vacancy networks, concerted hopping, rotational dynamics of polyanions, or disorder-induced flattening of the migration landscape. That explanatory power is

strategically important because it feeds back into descriptor design, transfer learning across ion chemistries, and the formulation of better generative objectives [9,13,87–107].

Still, ML potentials introduce their own scientific risks. Transferability is fragile when chemistry changes, when high-energy reactive configurations appear, or when interfaces generate compositions absent from the training set. Error bars on long-time diffusivity are difficult to quantify. Universal pretrained potentials are promising but cannot yet be assumed reliable for every alkali or multivalent solid-state battery chemistry without benchmarking [89,93,94,104]. Accordingly, the strongest workflows are those that treat ML potentials as adaptive scientific instruments rather than fixed black boxes: uncertainty-triggered retraining, targeted validation on transition states and interphases, and explicit comparison against DFT for the rare events that matter most.

Interfaces, imaging, and closed-loop discovery.

The bulk electrolyte has dominated the literature because it is the most tractable entry point for materials informatics. Yet practical all-solid-state batteries are frequently limited by interfaces, composite-electrode microstructure, and mechanically mediated degradation. Machine learning is now beginning to address these problems through several distinct routes [7–9,108–118,127–146]. One route is property prediction for interface-relevant quantities, such as elastic mismatch, decomposition products, or coating compatibility. A second is direct atomistic simulation of interface degradation with ML potentials, particularly in oxide and sulfide systems where cation interdiffusion and disordered interphases are hard to access by conventional approaches. A third is computer vision: segmentation and quantitative analysis of synchrotron tomography and other operando imaging modalities to detect lithium plating, voiding, or morphological evolution in solid-state cells [119].

The imaging direction is notable because it shifts machine learning from “predicting a material before synthesis” to “learning from a battery while it operates.” Deep segmentation of low-contrast tomography enables volumetric statistics on dendrites, pits, redeposition, and dead lithium that would be infeasible by hand, thereby transforming expensive operando measurements into structured training data for degradation-aware discovery [119]. That same logic extends to closed-loop control, where machine learning models classify battery state and reinforcement learning adjusts cycling strategy in real time. Although such work is closer to intelligent experimentation than to autonomous materials design in the narrow sense, it is directly relevant to discovery because it provides faster feedback on which interface or processing choices prolong life under realistic operating conditions [120–126].

Closed-loop optimization also points toward a broader methodological future: machine learning will likely be most valuable not when it replaces experiment, but when it allocates experimental budget. Bayesian or reinforcement-learning loops can decide which synthesis, composition, dopant level, coating, or formation protocol to test next; literature models can suggest constraints; surrogate models can rank remaining uncertainty; atomistic models can serve as mechanistic adjudicators [51–56,87–107,120–126,141,142]. In that sense, the mature frontier of machine learning for solid-state battery discovery is a systems problem rather than a single-model problem.

Challenges and Prospects

The principal scientific challenge is that the practical target is not a single material property but a coupled feasibility region. A high conductivity value by itself is not enough. The material must also resist reduction or oxidation, suppress electron leakage, maintain contact with electrodes, tolerate processing, and remain

compatible with the chosen cathode, anode, and stack-pressure regime. In machine-learning terms, the field is moving from single-objective supervised learning toward constrained, partially observed, multi-objective optimization [7–9,16,108–118,127–150]. This is why many conductivity models look better academically than they perform strategically: they are trained on labels that are too narrow relative to the actual decision boundary.

The second challenge is data semantics. Solid-state battery labels are not merely noisy; they are heterogeneous in meaning. “Ionic conductivity” may refer to bulk, grain-boundary, total pellet, or temperature-extrapolated conductivity; “electrochemical window” may reflect thermodynamic decomposition, kinetic passivation, or an interface-specific practical operating limit; “stability” may refer to chemical compatibility, cycling endurance, phase retention, moisture tolerance, or mechanical robustness. Unless future datasets encode these distinctions explicitly, model scaling alone will not solve the problem [8–10,14–16,45–52,108–118,128–150].

A third challenge is extrapolation beyond lithium-centric chemical space. The literature is still dominated by Li conductors, followed by a smaller but important body of work on Na systems. Multivalent conductors remain data-poor and physically distinct. Migration in Mg, Ca, Zn, or Al conductors is governed by stronger electrostatic interactions, more severe lattice distortions, and often different defect chemistries. Transfer learning from lithium is a useful starting point, but it cannot substitute for mechanistically grounded data generation and benchmarking [9,14–16,22,32,37,74–86]. The same problem appears when models trained on idealized crystalline structures are asked to reason about disorder, glassy phases, composite morphologies, or interphases.

These challenges motivate several concrete prospects. First, the community needs benchmark suites that separate family interpolation from true extrapolation, distinguish intrinsic from extrinsic labels, and evaluate uncertainty calibration alongside conventional error metrics. Second, multi-fidelity modeling should become standard: composition models for coarse search, graph models for structure-sensitive ranking, ML potentials for dynamic validation, and targeted experiments for final adjudication. Third, synthesis-aware discovery should be elevated from an afterthought to a first-class objective, using literature extraction, failed-experiment data, and process metadata rather than composition labels alone [45–56,87–107,141–148]. Fourth, interface-aware machine learning—combining reactive simulation, operando imaging, and closed-loop cycling—should be treated as central rather than peripheral, because the next bottleneck in solid-state batteries is increasingly interfacial rather than bulk [108–126,127–146].

More speculatively, the field appears ready for domain-specific foundation models that are truly multimodal: they would ingest crystal structures, compositions, diffusion trajectories, tomography volumes, electrochemical time series, and article text within a common latent space. Such models could support retrieval, hypothesis generation, uncertainty-aware optimization, and experiment planning. But a productive version of that agenda will require stronger curation norms, provenance tracking, and community agreement on what counts as a valid benchmark for scientific usefulness, not merely statistical fit [42,45–56,62,68,69,141,143,146].

Conclusion

Machine learning has already altered the practice of solid-state battery materials discovery. It has accelerated the search for inorganic solid electrolytes, enabled increasingly realistic atomistic simulations of ion transport and degradation, improved the analysis of complex operando data, and begun to close the

loop between prediction and experiment. The strongest achievements to date lie in bulk electrolyte screening and in the acceleration of atomistic transport studies. The hardest unsolved problems lie at interfaces, in synthesis translation, and in cell-level multi-objective optimization [17–37,57–107,108–128].

The central lesson of the literature is that no single model family is sufficient. Classical descriptor models remain useful in small-data regimes; graph and transformer architectures broaden representation power; generative models expand chemical exploration; ML potentials recover mechanism; and literature- and image-based models turn previously inaccessible information into structured evidence. The next phase of progress will likely come from composing these strengths rather than choosing among them. A mature discovery stack for solid-state batteries should couple curated and literature-mined data, uncertainty-aware screening, mechanistic simulation, interface-specific validation, and experiment-in-the-loop decision-making. If that integration succeeds, machine learning will not merely accelerate familiar searches; it will change the scale, scope, and epistemology of how solid-state battery materials are discovered.

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